Descriptive Analytics Report: YouTube Music

Streaming Performance, Genre Mix & Audience Behaviour Across 60 Songs

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Period Covered: February 2026 - July 2026

Note on data: This report uses a simulated but realistically structured YouTube Music catalogue of 60 songs x 6 months (22 variables per song), clearly labelled as simulated (no public song-level streaming API/dataset exists). Findings describe this 60-song sample only and are not claimed to generalise to the platform as a whole.

1. Objective

Build a descriptive analytics layer over the YouTube Music song-catalogue data to answer the following recurring business questions:

- How are songs distributed across genres, and which genres dominate total listening volume?
- Which genres carry the highest average plays per song, and how does that compare with song count?
- How is total platform listening trending month over month, and is growth broad-based or concentrated?
- Which songs are the most-played, and what does the shape of the play-count distribution look like?
- What is the overall audience composition by age group, device, and language?
- How does skip rate behave, and is it related to how many times a song is played?
- Which genres are popular with which age groups and in which regions?
- Which songs show viral-spike behaviour, and how does that differ from steady growth or decline?
- What does an overall catalogue-health summary look like across engagement, plays, and skip rate?

2. How This Would Work at Real Scale

Processing a large, constantly-updating song catalogue and its streaming events is an engineering problem before it is an analytics problem. A realistic architecture would ingest per-stream events from the platform, aggregate them by song, genre and time window, store raw and processed data, and expose the resulting KPIs through a data-visualisation layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the streaming dataset	Google, Claude & ChatGPT
Stream processing	Aggregate plays by song, genre, age group and device in near real time	Jupyter Notebook (Python)
Storage	Retain raw and processed monthly play/engagement data	Pandas / CSV / data warehouse
Batch aggregation	Monthly KPI rollups, trend labels and summary metrics	Jupyter Notebook (NumPy/pandas)
Report	Present analysis, charts and insights	Matplotlib / Seaborn / MS Word

3. Genre-Wise Song Distribution

Across the supplied catalogue, the dataset records 60 songs spanning 8 genres. The chart below shows how the song assortment is distributed across genres.

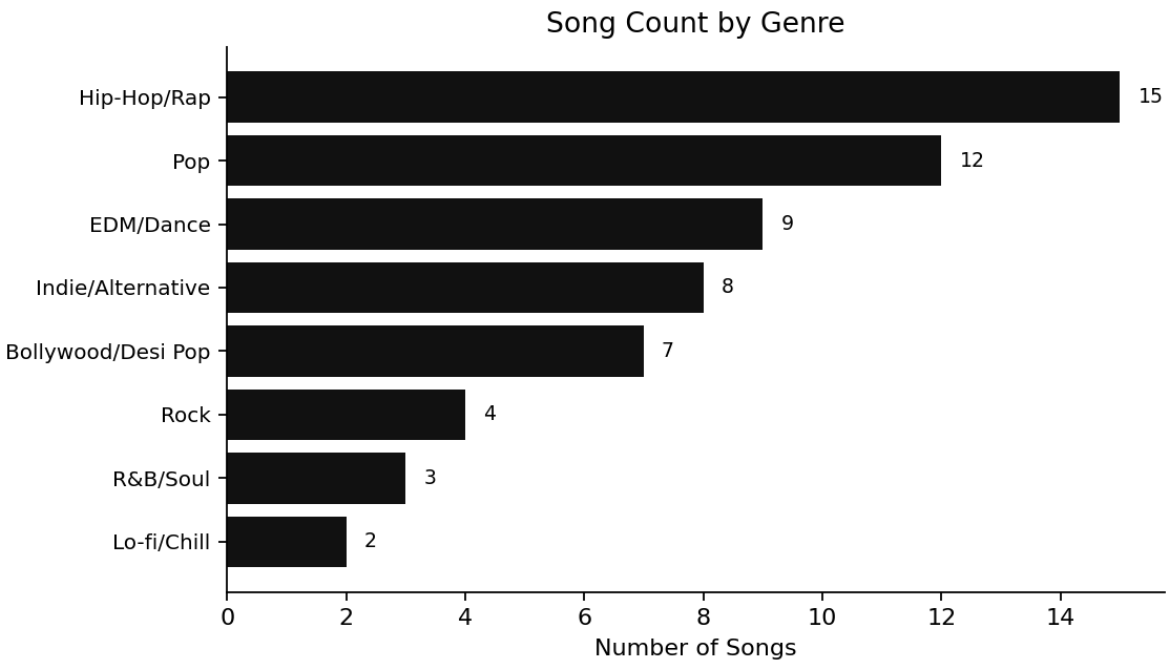


Figure 1. Song count by genre

INSIGHTS:

- Hip-Hop/Rap is the largest genre by song count, with 15 songs (25.0% of the catalogue), followed by Pop with 12 songs (20.0%).
- R&B/Soul and Lo-fi/Chill have the smallest assortments, with 3 and 2 songs respectively.
- The catalogue is weighted toward Hip-Hop/Rap, Pop and EDM/Dance by song count, but (as Section 4 shows) song count does not track play volume.
- Genre-level song counts can help guide catalogue-expansion and playlist-curation decisions.

KPI:
Total songs: 60
Total genres: 8
Largest genre: Hip-Hop/Rap (15 songs, 25.0%)
Smallest genre: Lo-fi/Chill (2 songs, 3.3%)

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4. Average Plays per Song by Genre

The overall average total plays per song across the supplied catalogue is ~359,200. The chart below shows how average plays per song varies by genre.

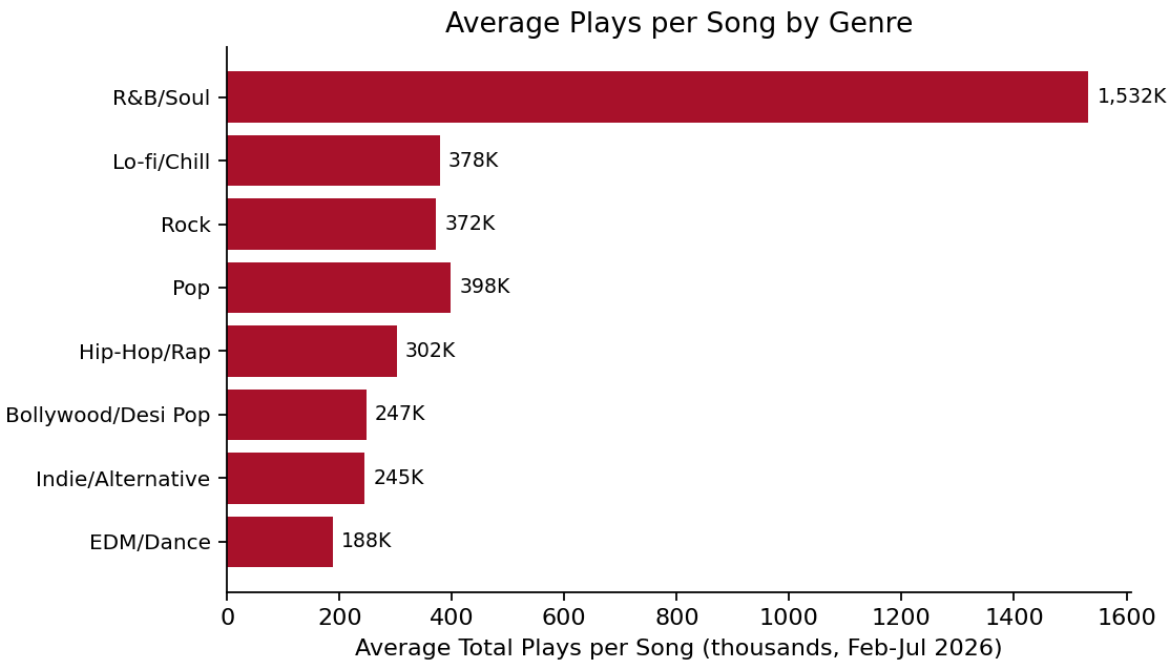


Figure 2. Average plays per song by genre

INSIGHTS:

- R&B/Soul carries by far the highest average plays per song (~1.53M), driven almost entirely by two viral-spike outlier tracks in that genre.
- Lo-fi/Chill, Rock and Pop follow at a much lower ~370K-400K average plays per song.
- EDM/Dance has the lowest average plays per song (~188K) despite being a mid-sized genre by song count.
- Average-per-song figures should be read alongside song count and spread (Section 14), since a genre average can be skewed by one or two breakout tracks.

KPI:
Overall avg. plays per song: ≈ 359.2K
Overall median plays per song: ≈ 228.6K
Highest-average genre: R&B/Soul (≈ 1.53M)
Lowest-average genre: EDM/Dance (≈ 188K)

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5. Share of Total Plays by Genre

The overall catalogue recorded 21,550,300 plays (21.55M) across the six-month window. The chart below shows how that volume is distributed across genres.

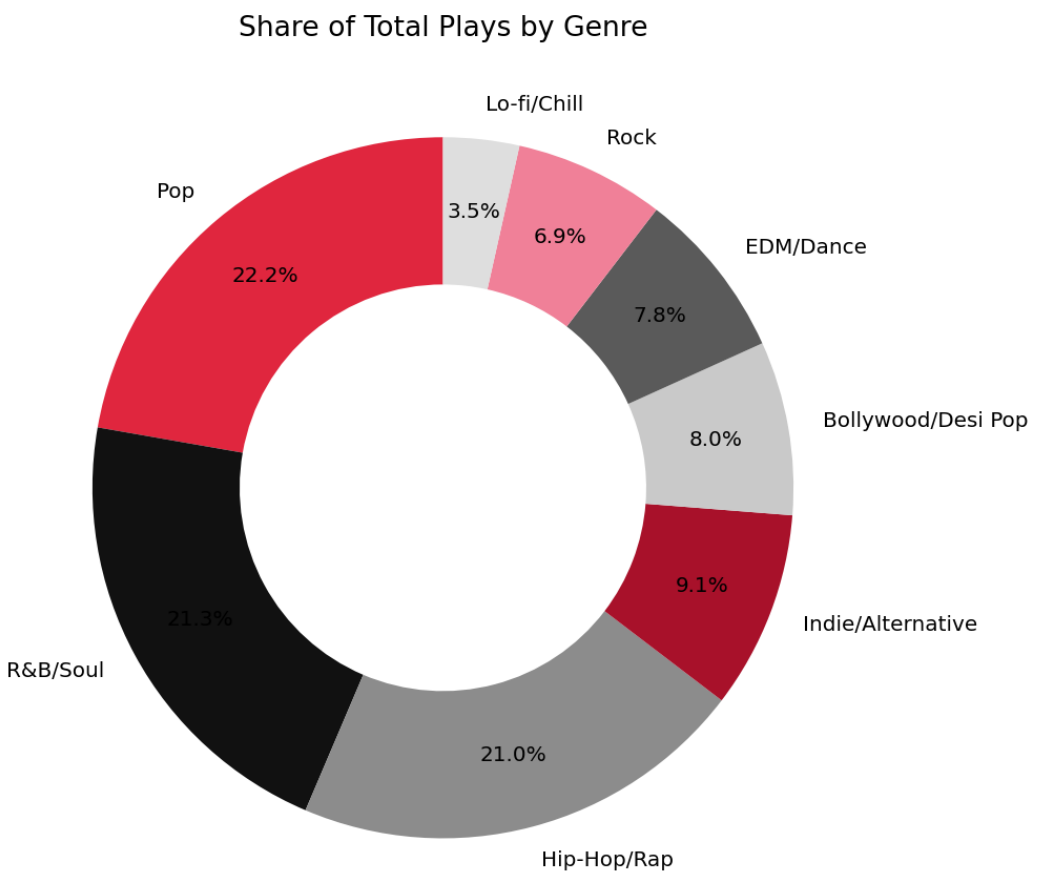


Figure 3. Share of total plays by genre

INSIGHTS:

- Pop (22.2%), R&B/Soul (21.3%) and Hip-Hop/Rap (21.0%) together account for 64.5% of all plays, despite representing only 50% of the songs in the catalogue.
- R&B/Soul's outsized 21.3% share from only 3 songs is driven almost entirely by two viral-spike outlier tracks — a reminder that composition and volume are not the same thing.
- The remaining five genres (Indie/Alternative, Bollywood/Desi Pop, EDM/Dance, Rock, Lo-fi/Chill) together share the remaining 35.5% of plays.

KPI:

Total plays (6 months): 21,550.3K (21.55M)
Top-3 genre share: Pop + R&B/Soul + Hip-Hop/Rap = 64.5%
Top-3 genre song share: 30 of 60 songs (50.0%)

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6. Platform-Wide Monthly Play Trend

Combining the 360 individual monthly-play records (60 songs x 6 months) into platform totals shows how total listening volume moved over the study window.

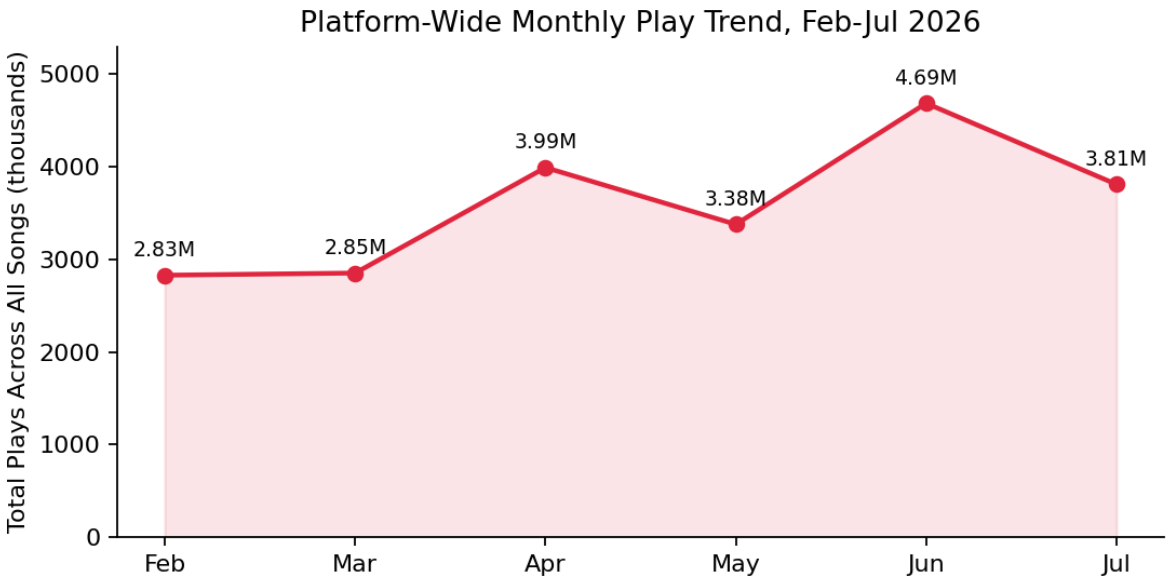


Figure 4. Platform-wide monthly play trend, February-July 2026

INSIGHTS:

- Total plays rose fairly steadily from 2.83M in February to a peak of 4.69M in June, before easing to 3.81M in July — a net +34.7% change from the first to the last month.
- June alone accounts for 21.7% of the six-month total, the single largest monthly share.
- Growth was not evenly distributed: roughly two-fifths of the catalogue (the 'stable' segment) moved only +3.3% over the period (see Section 9), so platform-level growth was concentrated in a minority of songs.

KPI:

Feb 2026 plays: 2,829.3K (13.1% of total)
Jun 2026 plays (peak): 4,686.8K (21.7% of total)
Jul 2026 plays: 3,810.9K (17.7% of total)
Net change, Feb→Jul: +34.7%

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7. Listener Age Group & Device Distribution

Viewed at the catalogue level, the majority of songs in the supplied dataset are primarily driven by young, mobile-first listeners.

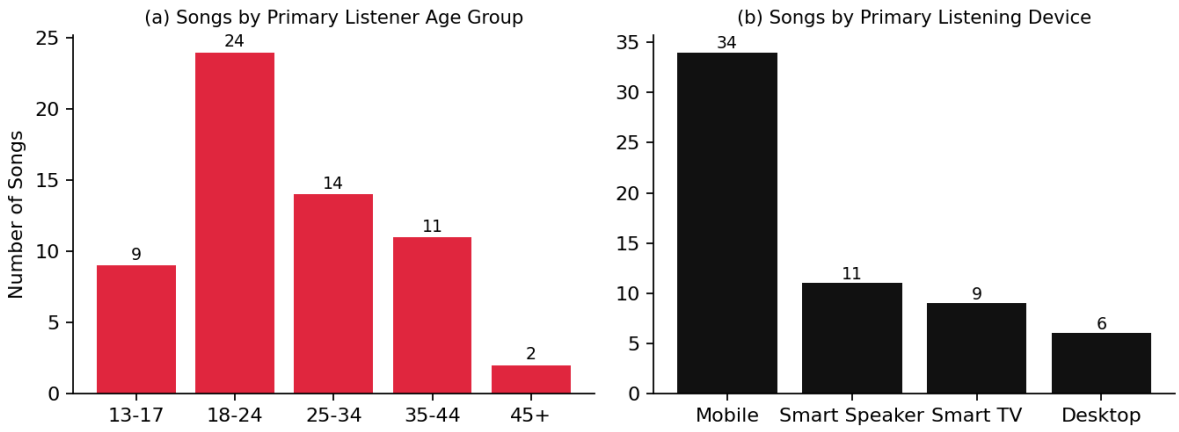


Figure 5. Frequency distribution of primary listener age group (left) and primary listening device (right)

INSIGHTS:

- Mobile is the primary listening device for 34 of 60 songs (56.7%) — more than the other three device categories (Smart Speaker, Smart TV, Desktop) combined.
- The 18-24 age bracket is the primary audience for 24 of 60 songs (40.0%), more than double the next-largest group (25-34, 23.3%).
- Only 2 songs (3.3%) are primarily driven by listeners aged 45+, confirming a catalogue skewed toward younger listeners.

KPI:

Mobile share: 34 of 60 songs (56.7%)
Largest age segment: 18-24 (24 songs, 40.0%)
Smallest age segment: 45+ (2 songs, 3.3%)

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8. Skip Rate Distribution

The histogram below shows how skip rate (%) is distributed across the full 60-song catalogue.

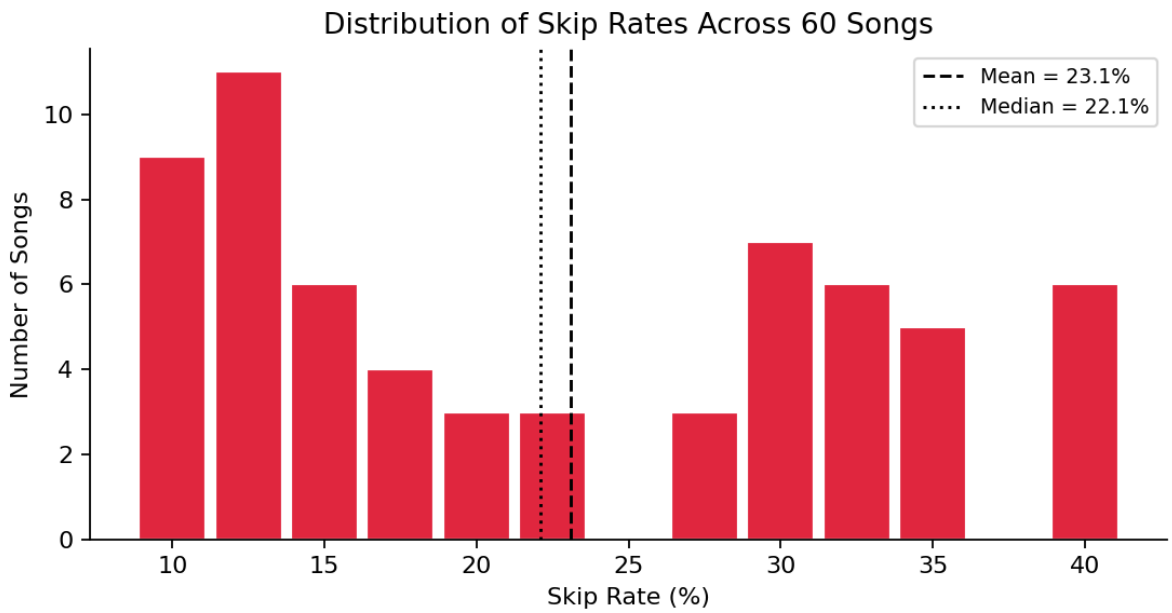


Figure 6. Distribution of skip rates across 60 songs

INSIGHTS:

- The distribution is roughly bell-shaped and centred near 22-23%, with no extreme outliers — unlike the play-count distribution, skip rate does not show a long right tail.
- The mean (23.1%) and median (22.1%) sit close together, indicating an evenly spread, less outlier-driven variable than total plays.
- Standard deviation is 10.6 percentage points, so most songs' skip rates fall roughly between 12.5% and 33.7%.

KPI:

Mean skip rate: 23.1%
Median skip rate: 22.1%
Std. deviation: 10.6 percentage points
Range: 8.5% - 41.5% (33.0-point spread)

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9. Trend-Type Segment Performance

Each song in the catalogue was internally labelled by its dominant six-month pattern (rising / falling / stable / viral-spike). The chart below compares the average Feb-to-Jul change for each segment.

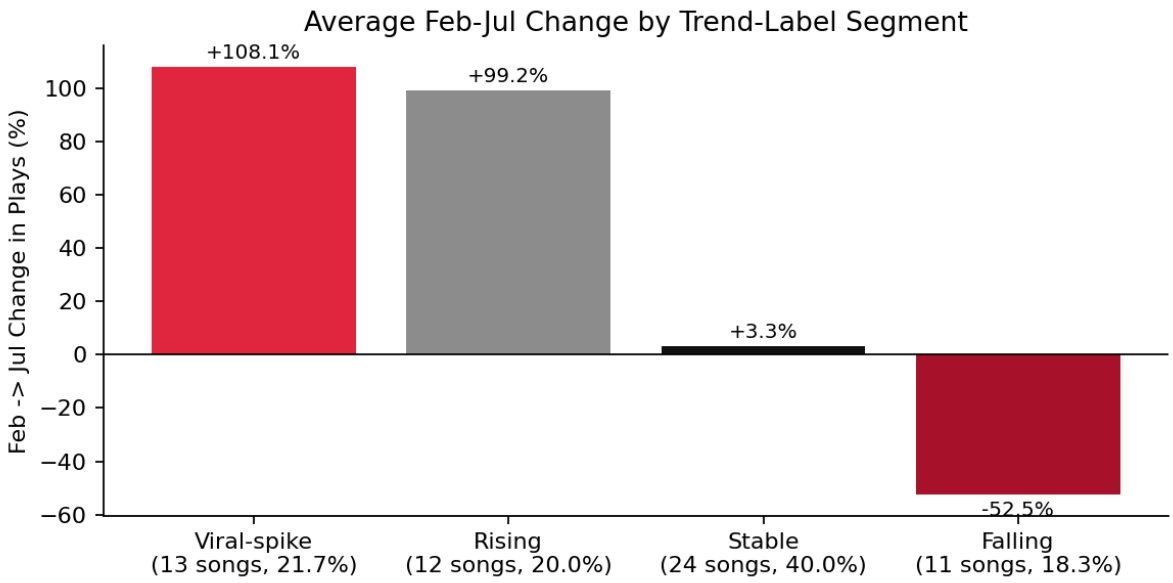


Figure 7. Average February-July play change by trend-label segment

INSIGHTS:

- Songs labelled 'viral-spike' (13 of 60, 21.7%) grew +108.1% on average from February to July; 'rising' songs (12 of 60, 20.0%) grew +99.2%.
- 'Falling' songs (11 of 60, 18.3%) declined -52.5%, while 'stable' songs — the largest single group at 24 of 60 (40.0%) — moved only +3.3%.
- Platform-level growth (Section 6) was driven almost entirely by the ~42% of songs that were either rising steadily or spiking virally; the stable 40% barely moved.

KPI:

Viral-spike songs: 13 (21.7%), avg. +108.1%
Rising songs: 12 (20.0%), avg. +99.2%
Stable songs: 24 (40.0%), avg. +3.3%
Falling songs: 11 (18.3%), avg. -52.5%

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10. Genre x Listener Age Group Cross-Tabulation

Cross-tabulating genre against primary listener age group (song counts) reveals which genres are popular with which audiences.

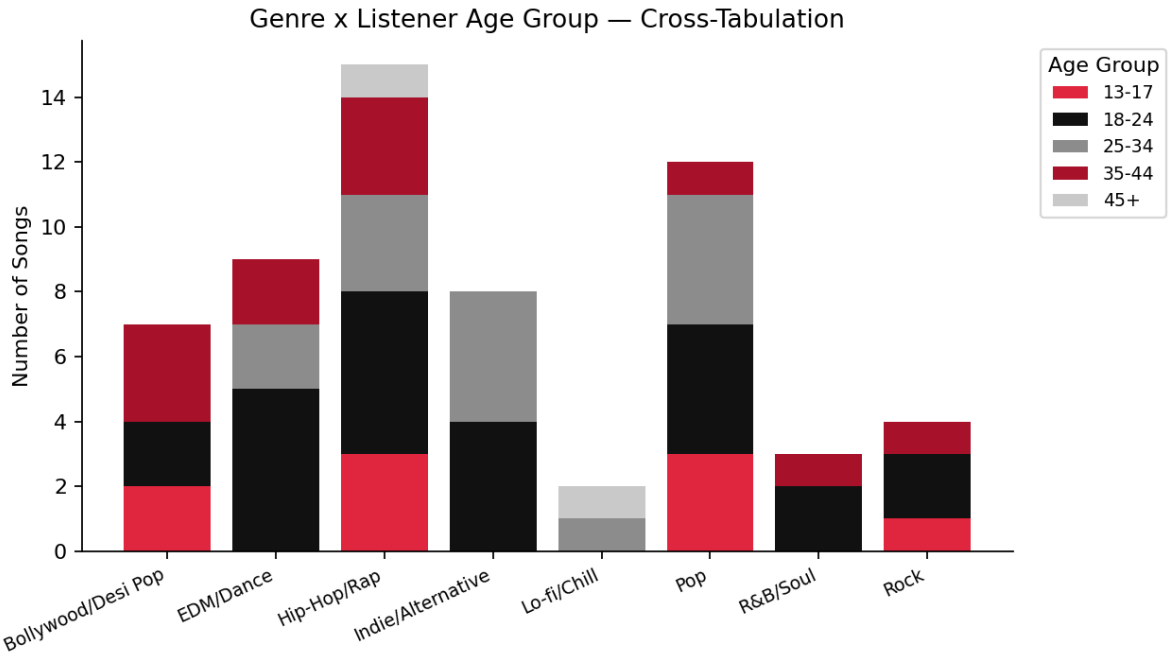


Figure 8. Genre x listener age group cross-tabulation

INSIGHTS:

- EDM/Dance and Indie/Alternative are driven almost entirely by the 18-34 bracket, with no songs in this sample primarily led by listeners aged 35 or above.
- Bollywood/Desi Pop shows its strongest concentration in the 35-44 bracket (3 of 7 songs), suggesting an older, likely diaspora-oriented core audience relative to the catalogue average.
- Hip-Hop/Rap and Pop show the broadest age spread of any genre in the catalogue, each reaching all five age brackets.

11. Genre x Region Play Volume

Cross-tabulating genre against top-performing region (aggregating total plays in thousands) shows where each genre's listening volume is concentrated.

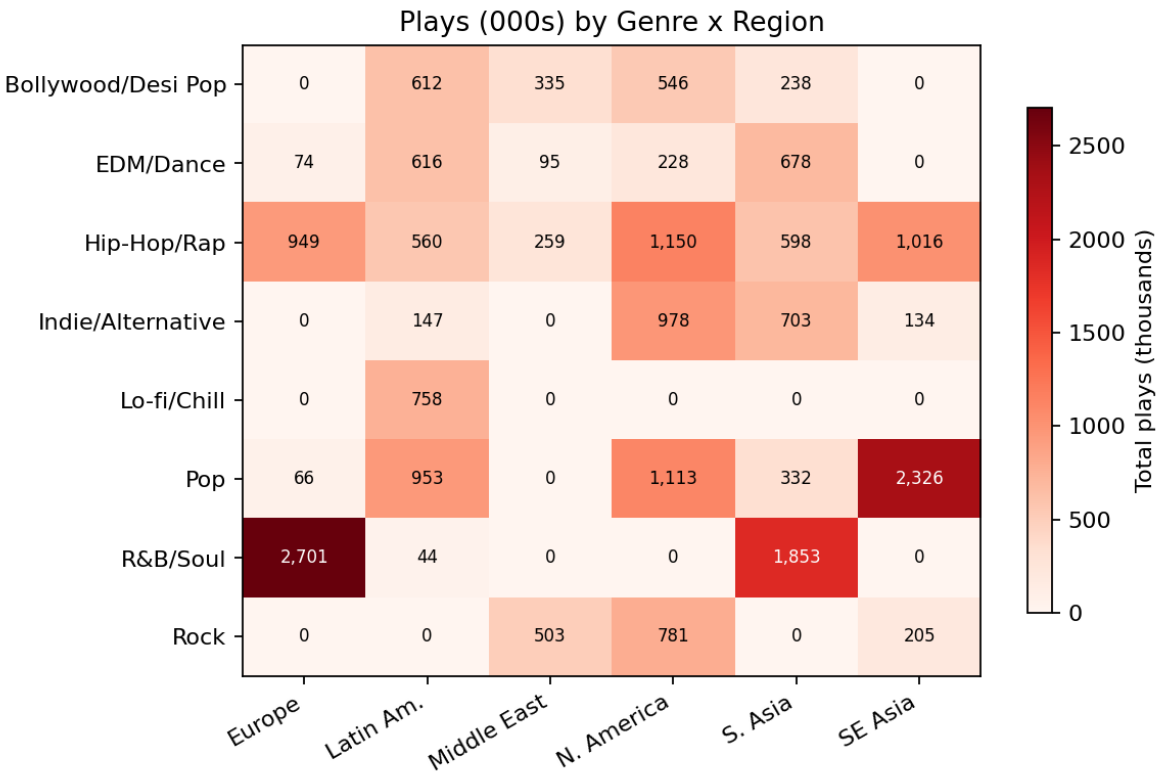


Figure 9. Total plays (thousands) by genre x region cross-tabulation

INSIGHTS:

- R&B/Soul in Europe (2,701K) is the single largest cell in the matrix, a direct reflection of one viral-spike track's primary region.
- Pop's play volume is heavily concentrated in Southeast Asia (2,326K of its 4,779K total, ~49%).
- Hip-Hop/Rap is the most geographically balanced genre in the catalogue, with meaningful play volume across five of the six regions rather than being concentrated in one or two.

12. Top 10 Most-Played Songs

The table and chart below list the ten songs with the highest six-month total plays in the supplied catalogue.

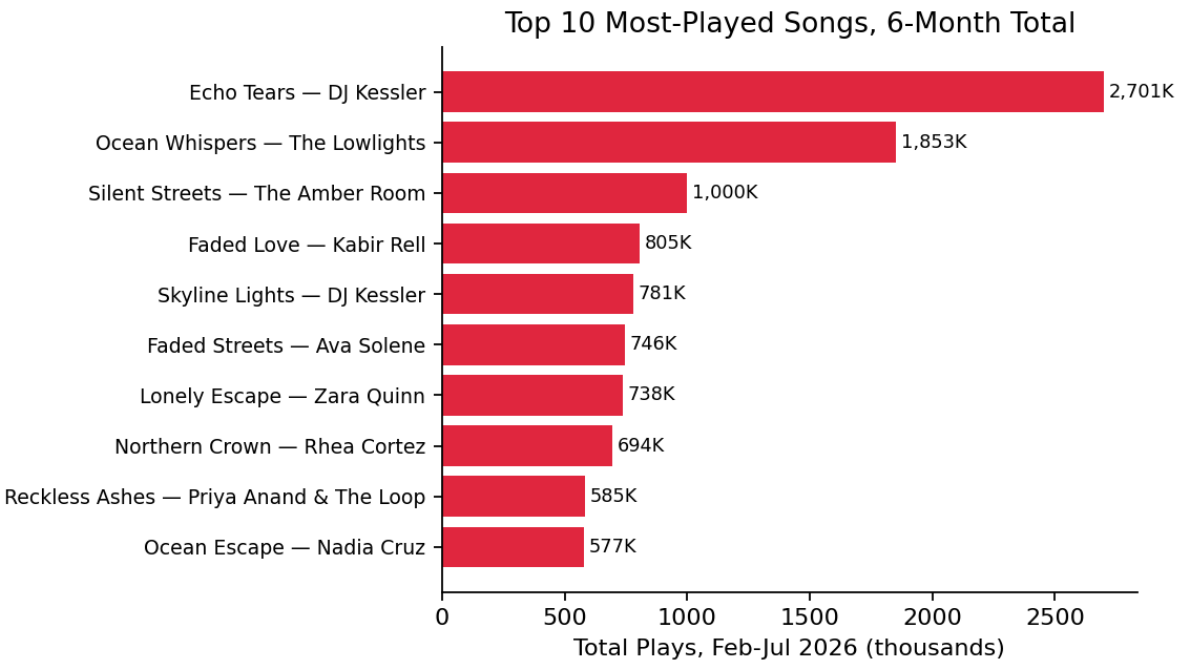


Figure 10. Top 10 most-played songs, 6-month total

Song	Artist	Genre	Total Plays (K)	Skip %
Echo Tears	DJ Kessler	R&B/Soul	2,700.7	33.1
Ocean Whispers	The Lowlights	R&B/Soul	1,852.6	31.8
Silent Streets	The Amber Room	Pop	999.7	39.3
Faded Love	Kabir Rell	Pop	804.6	13.0
Skyline Lights	DJ Kessler	Rock	781.0	28.0
Faded Streets	Ava Solene	Hip-Hop/Rap	746.0	14.8
Lonely Escape	Zara Quinn	Pop	737.8	30.2
Northern Crown	Rhea Cortez	Hip-Hop/Rap	694.1	39.8
Reckless Ashes	Priya Anand & The Loop	EDM/Dance	584.9	30.2
Ocean Escape	Nadia Cruz	Pop	576.6	33.1

INSIGHTS:

- 'Echo Tears' (2.70M plays) leads the catalogue by a wide margin, followed by 'Ocean Whispers' (1.85M) — together these two songs account for 21.1% of all plays despite being just 2 of 60 songs.
- Six of the top 10 songs belong to Pop, underlining Pop's strong showing despite ranking second (not first) in overall genre play share.
- The gap between rank 1 ('Echo Tears', 2.70M) and rank 3 ('Silent Streets', 1.00M) is larger than the gap between ranks 3 and 10 combined, consistent with the right-skewed distribution discussed in Section 14.

13. Monthly Trend — Top 5 Songs

Tracking the five highest-performing songs month by month shows how individual-song patterns differ from the platform-wide trend in Section 6.

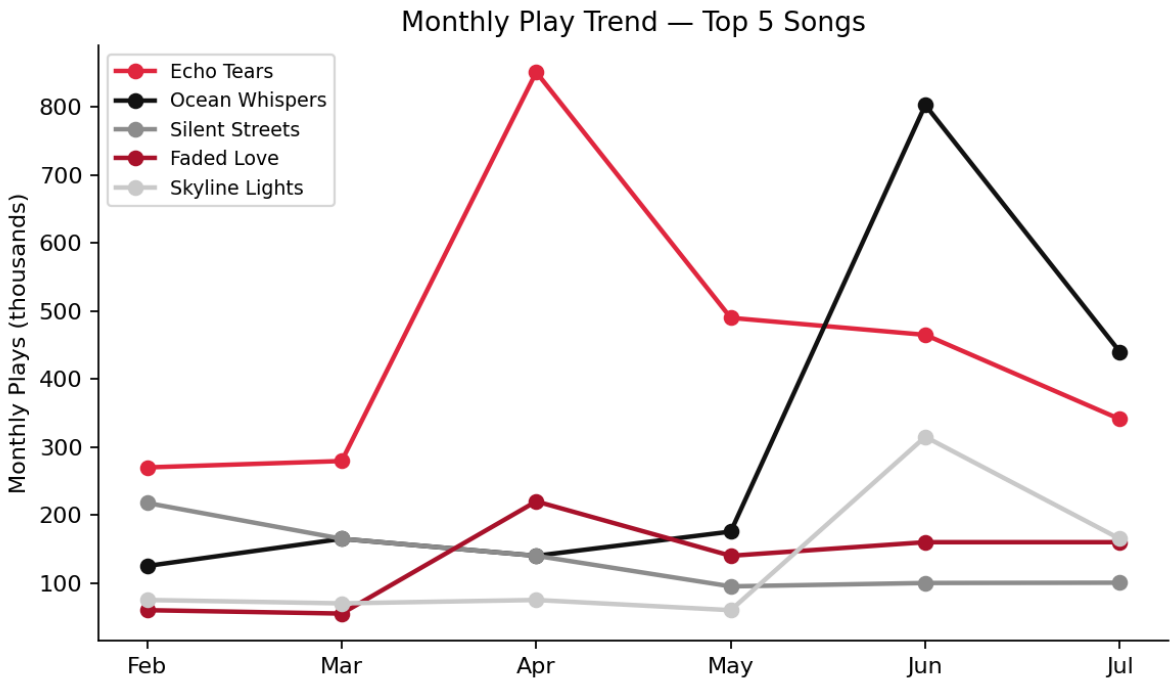


Figure 11. Monthly play trend for the five highest-performing songs

INSIGHTS:

- 'Echo Tears' shows the clearest viral-spike signature: plays jump from 279,400 in March to 851,400 in April (+205% month-on-month) before decaying over the following three months toward its pre-spike baseline.
- 'Ocean Whispers' shows the same pattern two months later, spiking from 175,600 in May to 803,100 in June (+357%) before falling by roughly half in July.
- 'Silent Streets', by contrast, shows a genuine, near-uninterrupted decline across all six months (217.7K to 100.4K, a 53.9% fall) — the classic signature of a song past its initial release window with no re-ignition event.

14. Spread of Total Plays by Genre

The box plot below (log scale, since raw values span two orders of magnitude) shows the spread of total plays within each genre.

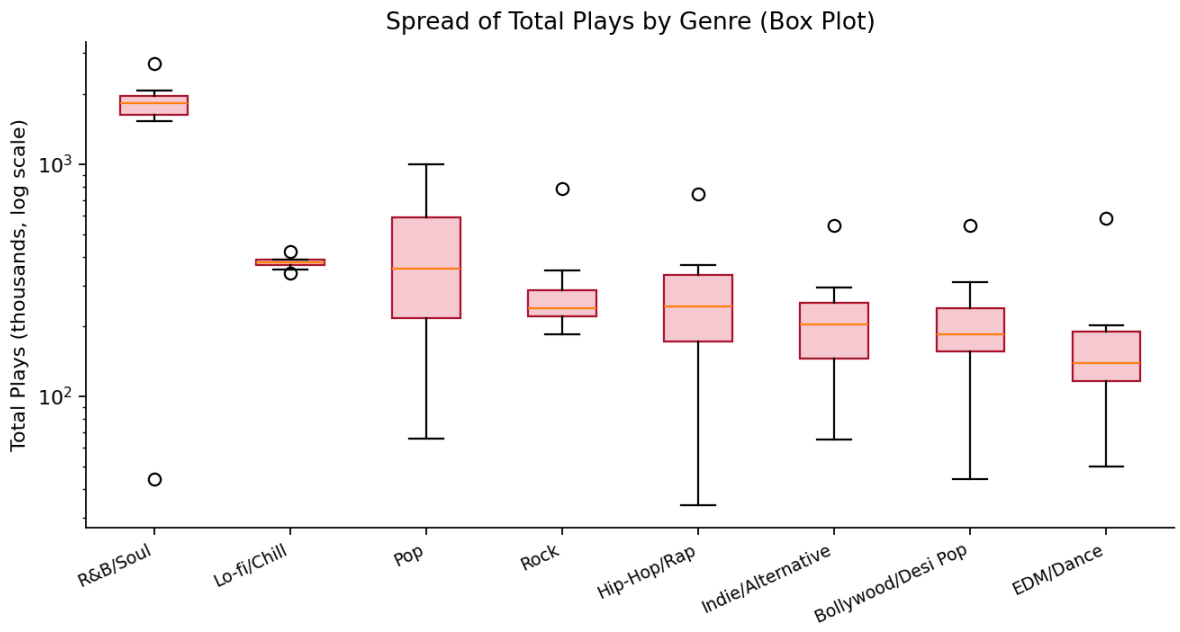


Figure 12. Spread of total plays by genre (box plot, log scale)

INSIGHTS:

- R&B/Soul shows the widest spread of any genre, driven by its two viral-spike outlier tracks; its whisker stretches far beyond every other genre.
- Rock and Bollywood/Desi Pop show tighter, more symmetric spreads, indicating more consistent, predictable performance within those genres.
- The standard deviation of total plays across the catalogue (429,500) is actually larger than the mean itself (359,200) — a strong statistical signature of a right-skewed, outlier-driven distribution.

KPI:

Total plays — range: 2,667.0K (33.7K - 2,700.7K)
Total plays — variance: 184,468.8
Total plays — std. deviation: 429.5K
Total plays — IQR: 273.6K

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15. Song Catalogue by Primary Language

The donut chart below shows how the 60-song catalogue splits across its five recorded primary languages.

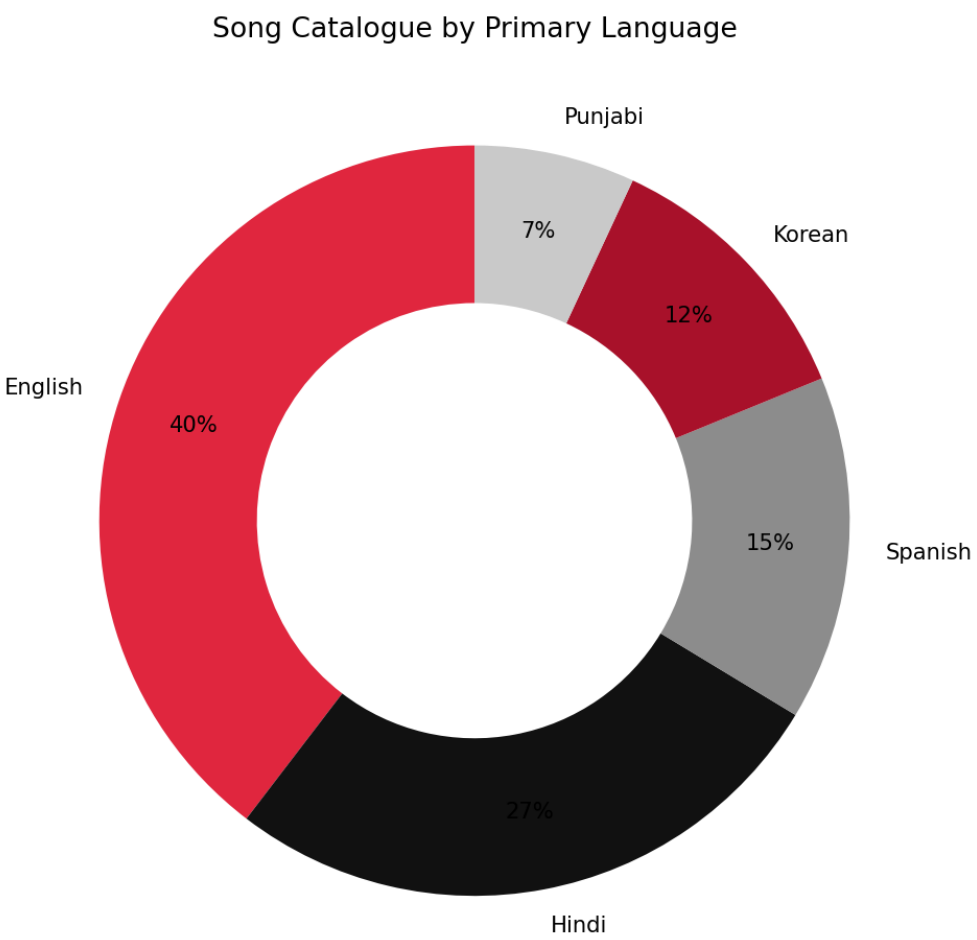


Figure 13. Song catalogue by primary language

INSIGHTS:

- English-language songs make up 40% of the catalogue, with Hindi (27%) as the second-largest language group — together these two languages account for two-thirds of all songs analysed.
- Spanish (15%), Korean (12%) and Punjabi (7%) make up the remaining third, reflecting a deliberately multi-language sample.

16. Relationship Between Plays and Skip Rate

The scatter plot below relates each song's total plays to its skip rate, using the 20-song raw-data sample in Appendix A, to test whether popularity and retention move together.

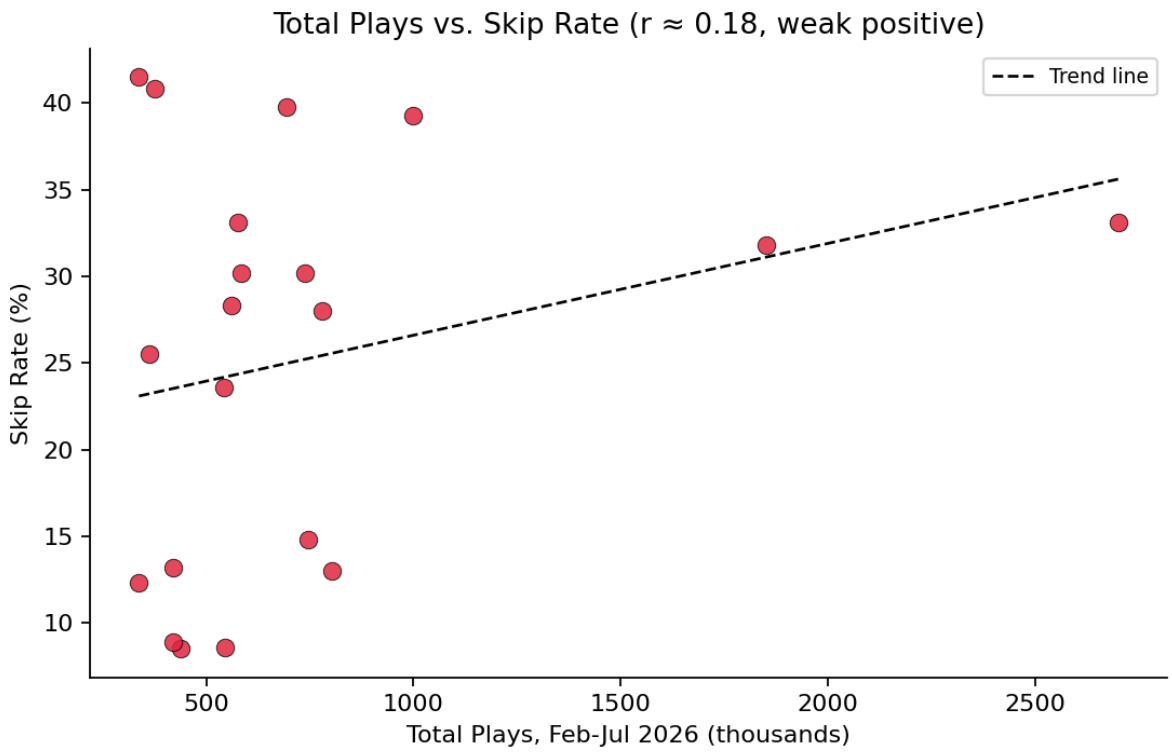


Figure 14. Total plays vs. skip rate, colored trend line shown

INSIGHTS:

- The correlation between total plays and skip rate across the catalogue is weak and slightly positive (r ≈ 0.18).
- A song's popularity (raw play count) says almost nothing about how well it retains listeners once they press play — for example, 'Faded Love' combines strong plays (804.6K) with a low 13.0% skip rate, while 'Electric Pulse' has under half the plays (376.5K) but a much higher 40.8% skip rate.
- Skip rate should therefore be tracked as an independent success metric, not inferred from play counts alone.

KPI:

Plays vs skip-rate correlation: r ≈ 0.18 (weak positive)

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17. Engagement Summary

Beyond raw plays, the catalogue also records likes, comments and shares — engagement metrics that describe how actively listeners responded, independent of play volume.

Metric	Value
Total plays (6 months)	21,550,300 (21.55M)
Total likes	1,352,645
Total comments	138,157
Total shares	308,847
Likes-to-plays ratio	≈ 6.3%

INSIGHTS:

- Likes track plays reasonably closely at the aggregate level (≈6.3% of plays convert to a like), consistent with likes being a low-friction engagement action.
- Comments and shares are a much smaller share of total plays, as expected for higher-effort engagement actions on a music-streaming platform.

18. Catalogue-Level Central Tendency & Spread Summary

Bringing together the central-tendency and spread statistics computed in this report gives a compact, catalogue-level statistical profile.

Variable	Mean	Median	Std. Dev.	Range
Total plays per song (K)	359.2	228.6	429.5	33.7 - 2,700.7
Skip rate (%)	23.1	22.1	10.6	8.5 - 41.5
Duration (minutes)	3.52	3.50	n/a	n/a

INSIGHTS:

- The gap between mean (359,200) and median (228,550) total plays per song is the key signal of the catalogue's long-tail structure: a small number of very high values pull the average upward.
- For a typical, middle-of-the-pack song, the median of ~228,600 plays is the more representative figure than the mean.

Summary of Key Metrics

Metric	Value
Total songs	60
Genres tracked	8
Languages tracked	5
Regions tracked	6
Total plays (6 months)	21,550,300 (21.55M)
Mean plays per song	≈ 359.2K
Median plays per song	≈ 228.6K
Feb→Jul platform-wide change	+34.7%
Mean skip rate	≈ 23.1%
Plays vs. skip-rate correlation	r ≈ 0.18
Mobile listening share	56.7% of songs
18-24 primary-audience share	40.0% of songs
Top-3 genre play share	64.5% (Pop, R&B/Soul, Hip-Hop/Rap)

Next Steps to Move from Descriptive to Predictive

- Streaming-volume forecasting by song and genre once a longer historical window is available.
- Viral-spike early-warning modelling using week-over-week play acceleration to flag likely spikes before they peak.
- Skip-rate-driven retention modelling linking skip rate to playlist placement and recommendation-engine outcomes.
- Regional promotion-spend optimisation once real transaction/marketing-spend data is layered onto the genre x region view (Section 11).
- Cohort analysis by release year and age group once longer historical catalogue snapshots become available.

This report answers what the supplied six-month catalogue shows. The next analytical layer would combine this streaming data with marketing, release-date and playlist-placement data to explain why songs spike or decline, and to forecast what is likely to happen next.

Glossary

Play count — the number of times a song was streamed in a given month on YouTube Music.

Skip rate (%) — the share of play sessions in which the listener skipped the track before it finished.

Trend label — an internal classification (rising / falling / stable / viral-spike) describing a song's overall monthly play pattern.

Primary age group / primary device / top region — the single dominant listener age bracket, device, and region associated with each song, used for song-level cross-tabulation.

Right-skew — a distribution shape where the mean sits above the median because a small number of high values pull the average upward; observed in total plays per song in this catalogue.

References

As the primary dataset is simulated, no external dataset citation is required for the core analysis. The following sources informed the general structure, statistical patterns, and industry context used to interpret the simulation, cited in APA style.

- YouTube Music. (2026). About YouTube Music. Google LLC.
- International Federation of the Phonographic Industry. (2025). IFPI Global Music Report 2025. IFPI.
- Luminate. (2025). Luminate 2025 Midyear Music Report. Luminate Data LLC.
- Course materials: Business Analytics — Descriptive Analytics module lecture notes and slides, [Instructor name], [Institution name], 2026.

Appendix A — Sample of Raw Dataset (20 of 60 Songs, Sorted by Total Plays)

ID	Title	Artist	Genre	Language	Total Plays (K)	Skip %
S029	Echo Tears	DJ Kessler	R&B/Soul	English	2,700.7	33.1
S024	Ocean Whispers	The Lowlights	R&B/Soul	English	1,852.6	31.8
S008	Silent Streets	The Amber Room	Pop	Korean	999.7	39.3
S016	Faded Love	Kabir Rell	Pop	English	804.6	13.0
S048	Skyline Lights	DJ Kessler	Rock	Spanish	781.0	28.0
S031	Faded Streets	Ava Solene	Hip-Hop/Rap	Korean	746.0	14.8
S042	Lonely Escape	Zara Quinn	Pop	Spanish	737.8	30.2
S011	Northern Crown	Rhea Cortez	Hip-Hop/Rap	English	694.1	39.8
S025	Reckless Ashes	Priya Anand & The Loop	EDM/Dance	English	584.9	30.2
S038	Ocean Escape	Nadia Cruz	Pop	English	576.6	33.1
S030	Sunset Bones	Nova Sterling	Hip-Hop/Rap	Korean	560.0	28.3
S007	Dream Sky	Kian Osei	Bollywood/Desi Pop	Spanish	545.8	8.6
S037	Faded Crown	Nadia Cruz	Indie/Alternative	Hindi	542.6	23.6

ID	Title	Artist	Genre	Language	Total Plays (K)	Skip %
S018	Neon Tide	Nova Sterling	Bollywood/Desi Pop	Hindi	439.0	8.5
S045	Electric Shadows	The Paper Suns	Hip-Hop/Rap	Spanish	421.4	8.9
S043	Lonely Streets	Ava Solene	Lo-fi/Chill	Punjabi	420.5	13.2
S053	Electric Pulse	Coastline	Pop	Hindi	376.5	40.8
S023	Rewind Star	The Amber Room	Indie/Alternative	Punjabi	364.4	25.5
S058	Static Whispers	Devon Rae	Indie/Alternative	English	338.3	12.3
S022	Static Signal	Jisoo Han	Lo-fi/Chill	Hindi	338.0	41.5